MYBOOKCATALOG MOBILE APPLICATION

Task 2: Bug Documentation and Resolution Report

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Application Name: MyBookCatalog
Platform: Android Studio (Java)
Language: Java & XML

Testing Summary

1. Functional Testing

The **MyBookCatalog** mobile application was thoroughly tested to ensure that all features functioned correctly according to the project requirements. The application launched successfully without any build errors or runtime exceptions. The home screen displayed the complete list of available books, allowing users to browse the catalogue with ease.

Each book item was tested by selecting it from the catalogue. The application correctly displayed the book details, including the title, author, description, cover image, and fixed purchase price. Navigation between the Home screen and the Book Details screen was smooth and responsive.

The search functionality was tested by entering different book titles and keywords into the search bar. The application successfully filtered the available books and displayed only the matching results, making it easy for users to locate the books they wanted to purchase.

The online purchasing feature was also tested successfully. Users were able to select a book, view its fixed price, click the **Buy Now** button, and complete the purchase process. After the purchase was completed, the application displayed a confirmation message indicating that the transaction had been processed successfully.

All buttons, menus, and navigation controls functioned correctly throughout the testing process, and no functional errors were identified.

Result: Passed

2. UI/UX Testing

The user interface of the application was tested to ensure that it was attractive, consistent, and user-friendly. All screens displayed correctly without layout issues, and the application maintained a clean and professional appearance.

The text was clear and readable, while the images were displayed correctly without distortion. Buttons responded immediately when pressed, and the navigation between activities was fast and seamless.

The search bar was positioned appropriately, making it simple for users to search for books. The purchase button was clearly visible, allowing users to complete purchases quickly and easily. The overall design provided a smooth and enjoyable user experience.

No user interface problems or usability issues were encountered during testing.

Result: Passed

3. Device Compatibility Testing

The application was tested on both an Android Emulator and a physical Android smartphone to verify compatibility across different Android devices.

The application installed successfully on both devices without installation errors. All features, including browsing books, searching for books, viewing book details, and purchasing books online at fixed prices, functioned correctly on each device.

Performance remained stable throughout the testing process. The application loaded quickly, navigation was responsive, and no crashes, freezing, or compatibility issues were observed.

Devices Tested:

- Android Emulator
- Android Smartphone

Result: Passed

4. Debugging

Debugging was performed throughout the development process using Android Studio development tools, including **Logcat**, **Build Output**, **Gradle Build**, and the **Android Debugger**.

The source code was reviewed carefully to identify and resolve syntax errors, logical errors, and runtime exceptions. The AndroidManifest.xml file was verified to ensure that all application activities were properly declared. XML layout files were checked for correct view IDs, constraints, and user interface consistency.

The RecyclerView adapter, search functionality, and purchase process were tested repeatedly to ensure reliable performance. Image resources were verified to ensure that every book cover displayed correctly.

After completing the debugging process, the project compiled successfully without any build errors or warnings. The application executed smoothly, and no unexpected behaviour was observed during testing.

Debugging Tools Used:

- Android Studio Logcat
- Build Output
- Android Debugger
- Gradle Build

Result: Passed

5. Final Resolution

Following comprehensive testing and debugging, all identified issues were successfully resolved. The MyBookCatalog application builds successfully in Android Studio without compilation errors and installs correctly on both the Android Emulator and a physical Android phone.

The application performs all intended functions efficiently. Users can browse the complete catalogue of books, search for specific books using the search feature, view detailed information about each book, and purchase books online at a fixed price. After a successful purchase, the application displays a confirmation message, ensuring that users receive feedback on their completed transactions.

The application is stable, reliable, responsive, and easy to use. During final testing, no runtime errors, crashes, performance issues, or compatibility problems were encountered. All project requirements were successfully achieved, and the application is ready for deployment.

Overall Status

All bugs were resolved successfully.

The MyBookCatalog mobile application successfully passed **Functional Testing**, **UI/UX Testing**, **Device Compatibility Testing**, and **Debugging**. The application runs smoothly on Android devices without any build errors, runtime errors, or compatibility issues.

Users can:

- Browse the complete book catalogue.
- Search for books by title using the search feature.
- View detailed information about each book.
- Purchase books online at a fixed price.
- Receive a purchase confirmation after completing a transaction.